

JESICA MARIA RAMIREZ TOSCANO

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PROFILE

- **Motivation:** Passionate about the responsible development of AI that is safe and beneficial to society.
- **Soft Skills:** Cross-functional collaborator, resourceful, and experienced in communicating technical findings to non-technical stakeholders.
- **Technical Skills:** Applied causal inference, including difference-in-differences, regression discontinuity, synthetic control, and instrumental variables; supervised machine learning; experimental design; statistics; and exploratory data analysis.
- **Programming Skills:** Python, R, SQL, C#, HTML, and JavaScript.

EDUCATION

University of Chicago

Master of Science in Computational Analysis and Public Policy

September 2019 – June 2021

Honors Distinction

Relevant Coursework: Causal Inference (2 courses), Machine Learning (2 courses), Databases, Data Visualization, and Computer Science Applications (2 courses)

Universidad de las Américas Puebla, Mexico

Bachelor's Degree in Economics

August 2012 – June 2017

Magna Cum Laude

First Place, Empirical Research Thesis Regional Competition in Puebla

July 2017

PROFESSIONAL EXPERIENCE

Uber Technologies Inc.

September 2021 – Present

Scientist II

October 2024 – Present

Earners Progression Science

- Built a causal estimation framework using order-level fixed-effects regression to quantify the marketplace cost of courier behavior, enabling economically grounded quality measurement and informing policy decisions on earner-quality metrics.
- Designed experiments and behavioral measurement frameworks informing earner-quality initiatives across Product, Engineering, and Operations; contributed to a real-time feedback system generating \$36M in annual cost savings.

Applied Scientist II

September 2021 – October 2024

Policy Research – Legal Data & Economics

- Developed multi-scenario economic analyses to support regulatory engagement with the New York Attorney General, translating complex platform data into externally auditable evidence for a government proceeding.
- Produced statistical analyses and data investigations for government regulators across multiple jurisdictions, translating internal platform data into externally auditable evidence packages.
- Developed a framework to maximize user privacy in regulatory data productions and built self-service visualization tools enabling Legal and Operations teams to independently access and interpret driver-activity data in arbitration and litigation matters.
- Designed and deployed a classification model to automate the detection of notifiable incidents for Australian regulators, reducing compliance-review burden with projected savings of AUD 500,000 per year; nominated for the 2023 Reimagine Award.

- Modeled litigation risk and financial exposure across regulatory matters, providing data-driven inputs to legal strategy and external negotiations.

Deep Dive
Data Scientist

June 2020 – August 2020

- Collaborated on the development of the company's natural-language processing pipeline, including designing functions and objects to process large-scale text data.

Banco de México (Central Bank)
Economist

August 2017 – August 2019
General Directorate of Economic Research

- Quantified the tone of Mexico's central-bank policy communications across more than 80 statements from 2008 to 2019 using natural-language processing, producing a policy-signal index that supported monetary-policy strategy and external communication decisions.
- Derived inflation expectations and interest-rate forecasts at horizons ranging from one month to three years using market instruments, including swaps and bonds, and survey data, directly supporting monetary policy decisions at the sovereign level.

PROJECTS

- **Classification and Prediction:** *COVID-19 in Mexico: Predicting Severe Disease Outcomes.* Used balanced and weighted random-forest models to predict individual deaths and hospitalizations using Mexican government data, demonstrating how public-health datasets can inform policy interventions.
- **Databases and Application Development:** *Mapping Crimes in Mexico City.* Built a web application providing real-time crime-risk information by location and time of day, using Google's API and more than two million observations from the city government's open-data API.
- **Neural Networks and Social Polarization:** *Predicting Polarization from News Outlet Tweets.* Built a convolutional neural network in PyTorch to classify the social polarization of Twitter comments, contributing to research on measuring online discourse for policy audiences.
- **Web Scraping and Time-Series Prediction:** *Hohonu: Water-Level Monitoring.* Collaborated on a data pipeline to clean, calibrate, and predict local water-level data; scraped more than 200 URLs to obtain standardized datum measurements and optimized vector autoregression predictions using external data.